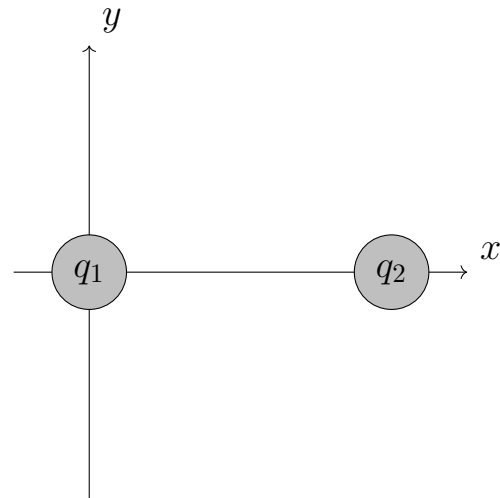

A charge of $q_1 = +40 \text{ nC}$ sits 5 mm away from a charge of $q_2 = -50 \text{ nC}$.
(*Hint:* $1 \text{ nC} = 10^{-9} \text{ C}$)

- (a) Draw the electric field lines.
- (b) What is the magnitude and direction of the electric field at a location halfway between the two charges?
- (c) What is the magnitude and direction of the electric field at $x = 8 \text{ mm}$?



A charge of $q_1 = +40 \text{ nC}$ sits 5 mm away from a charge of $q_2 = -50 \text{ nC}$.
(*Hint:* $1 \text{ nC} = 10^{-9} \text{ C}$)

- (a) Draw the electric field lines.
- (b) What is the magnitude and direction of the electric field at a location halfway between the two charges?
- (c) What is the magnitude and direction of the electric field at $x = 8 \text{ mm}$?

